

Weixiang Guo

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Educational Background

Nanyang Technological University (MSc (Eng.))

Aug 2025 – Present

- School: Electrical and Electronic Engineering
- Major: Electronics Technology
- Dissertation: Multimodal UAV Perception for Through-Glass Human Activity Understanding in Urban Environments

Taiyuan University of Technology (B.Eng.)

Sept 2021 – Jun 2025

- School: Electronic Information Engineering
- Major: Electronic Information Engineering
- GPA: 89.77/100 (Class Ranking: 3/35; Department Ranking: 6/70)

Academic Experience

Research Student — NTU Internet of Things Lab & Center for Advanced Robotics Technology and Innovation (CARTIN)

Jun 2025 – Present

- **Supervisors:** Professor Lihua Xie, Senior Research Fellow Shenghai Yuan
- **Research Focus:** Conducted research on multi-robot collaboration, perception, and energy-aware planning. Developed STARC, a real-time cross-LiDAR AR framework for human-robot situational awareness; proposed PERAL, a perception-aware motion control scheme for spherical robots; and implemented an energy-constrained SE(2) planner for hybrid RTG-solar rovers
- **Achievements:** Three preprints on multi-robot cross-LiDAR collaboration (STARC), perception-aware motion control (PERAL), and energy-constrained rover planning [J1-J3]

Research Assistant — Shanxi Provincial Key Laboratory for Data Governance and Intelligent Decision

Jan 2024 – Sept 2024

- **Supervisor:** Professor Jumin Zhao, Professor Dengao Li
- **Responsibilities:** Assisted in the MQAM Backscatter Modulation and Cross-Protocol Communication project by optimizing impedance matching, designing RF mixer modules, a 50 MHz low-pass filter, and an FPGA-based programmable voltage source. Contributed to the STM32-based LoRa Remote Networking Test Box by designing and programming the control board and implementing multi-sensor data transmission

Undergraduate Researcher — Electronic Information Innovation Laboratory, Taiyuan University of Technology

Jan 2023 – Dec 2023

- **Supervisors:** Associate Professor Bo Zhang, Associate Professor Yanlong Liu
- **Position:** Member of the Embedded Systems Group
- **Responsibilities:** Developed embedded control and sensing systems integrating STM32 and Arduino platforms for autonomous tracking and environmental perception. Designed multi-module controllers covering motion actuation, visual recognition, and wireless communication, and contributed to a patented CPU chip design for side-channel attack resistance
- **Achievements:** Received Second Prize (Undergraduate Group, Shanxi Division) in the 2023 National Undergraduate Electronic Design Contest [A1], and one invention patent under review [P1]

Publications

[†] Equal contribution. All works are preprints / under review.

[J1] STARC: See-Through-Wall Augmented Reality Framework for Human-Robot Collaboration in Emergency Response

Shenghai Yuan[†], Weixiang Guo[†], Tianxin Hu, Yu Yang, Jinyu Chen, Rui Qian, Zhongyuan Liu, Lihua Xie
Preprint: arXiv:2509.15507

[J2] Energy-Constrained Navigation for Planetary Rovers under Hybrid RTG-Solar Power

Tianxin Hu[†], Weixiang Guo[†], Ruimeng Liu, Xinhang Xu, Rui Qian, Jinyu Chen, Shenghai Yuan, Lihua Xie
Preprint: arXiv:2509.15062

[J3] **PERAL: Perception-Aware Motion Control for Passive LiDAR Excitation in Spherical Robots**
Shenghai Yuan[†], Jason Wai Hao Yee[†], **Weixiang Guo**, Zhongyuan Liu, Thien-Minh Nguyen, Lihua Xie
Preprint: arXiv:2509.14915

Patents

[P1] CPU Chip Resistant to Power Side-Channel Attacks Based on Energy Trace Balancing	Pending
Qitian Yan [†] , Shaojie Yao [†] , Zhengbo Ning [†] , Jiajia Lu [†] , Weixiang Guo [†]	
Application No.: CN202510058591.6, <i>National Intellectual Property Administration, P. R. China</i>	Jan 2025

Awards

[A1] (Provincial Level) Second Prize in the Shanxi Division of the 2023 National Undergraduate Electronic Design Contest	Aug 2023
<i>Organizing Committee of the National Undergraduate Electronic Design Contest, Ministry of Education of the PRC</i>	
[A2] (Provincial Level) Second Prize in Group A of the 2024 Shanxi Division of the “Datang Cup” National College Students’ New Generation ICT Competition (Information and Communication Engineering Practice Track)	May 2024
<i>Organizing Committee of the “Datang Cup” ICT Innovation Competition</i>	
[A3] (University Level) Triple-A Student of the College of Electronic Information and Optical Engineering	Dec 2023
<i>Taiyuan University of Technology</i>	
[A4] (University Level) Outstanding Student Scholarships (Top 5%) — awarded six times from 2021 to 2024	2021–2024
<i>Taiyuan University of Technology</i>	

Technical Expertise

Programming: C, C++, Python, basic CUDA

Robotics & Simulation: ROS/ROS2, Gazebo Classic & GZ Sim, RViz, URDF/SDF modeling

Embedded Systems: NVIDIA Jetson (Orin AGX, Orin NX), STM32; Linux

Perception & Fusion: SLAM, OpenCV, PCL, deep learning frameworks (PyTorch, TensorFlow)

Hardware Design: Altium Designer; multilayer PCB design and soldering